API DOCUMENTATION

(APPLICATION PROGRAMMING INTERFACE)

FOR PAYKADUNA 3RD PARTY EXPERT SYSTEM INTEGRATIONS

VERSION 8

PRESENTED BY

PRIMEGUAGE SOLUTIONS

Document Revision History

Date	*Version*	*Reason for Change*	*Author*
08/09/2025	08	1. An update to section 2.9 consists of the Year of Assessment endpoint. 2. An addition of a generic endpoint for any TSP to include additional information in section 2.10.	Primeguage Solutions
02/04/2025	07	1. Update on section 2.4 Create a Payment Transaction. 2. Update on section 2.8 Update Payment Redirect URL	Primeguage Solutions
10/1/2025	06	1. Update on sections 2.1 and 2.2. 2. Update the bill creation endpoint to pass taxpayer name and details.	Primeguage Solutions
21/10/2024	05	1. TaxstationId is required and (LgaId no longer required) for TaxPayerRegistration Endpoint	Primeguage Solutions
26/09/2024	04	Update on Tax Offices	Primeguage Solutions
24/09/2024	03	1. Updated the GetinvoiceURL 2. Updated for Remita Payment	Primeguage Solutions
11/09/2024	02	**1.** ***Updated Create Bill Endpoint:*** Now accepts the "TSP EngineCode" and allows the "Identifier" to be either the Taxpayer ID from the legacy system or the TPUI from PayKaduna.com. Both identifiers can be used interchangeably. **2.** ***Registration Endpoint:*** Adjusted to fit unique peculiarities of all TSPs, now only requiring "FirstName" and "LastName" fields for registration. Other fields have been relaxed.	Primeguage Solutions
20/06/2024	01	Initial Copy	Primeguage Solutions

1.0 INTRODUCTION

1.1 OBJECTIVES

The objective of this document is to define the interface between the PayKaduna Intelligent Billing System and third-party platforms for the purpose of carrying out specific functions. The communication mode is via REST web

service and is accessible via the URL:

Testing: https://testingapi.paykaduna.com/

Production: https://api.paykaduna.com/

1.2 INTENDED AUDIENCE

This document is intended to serve as specifications for the technical team of an integrating party to design and develop components to interface with PayKaduna IBS.

1.3 SECURITY & ACCESS

Step 1: Primeguage shares the API key with the integrating party

Step 2: When the integrating party wants to make a request:

1. For [POST] REQUESTS:

   Compute a HMAC SHA256 hash (Base64) of the JSON payload, using the API key as the secret key. This is your digital signature

2. For [GET] requests:
   Compute a HMAC SHA256 hash (Base64) of the API path and query string, using the API key as the secret key. This is your digital signature e.g. if making a GET request to https://testingapi.paykaduna.com/api?param=23, you will hash /api?param=23

   N.B: for testing purposes only you may use an online tool like https://www.devglan.com/online-tools/hmac-sha256-online to create the hash, make sure the JSON payload is formatted to be single line, again for testing purposed only an online tool like https://www.text-utils.com/json-formatter/ can be used to achieve this.

   Take note of the JSON formatting when building your internal routines.

Step 3: Set the **X-Api-Signature** header to your digital signature

Step 4: Attach the X-Api-Signature to your request

Step 5: Start Consuming PayKaduna web services as described below

N.B: The Api key will be reset periodically for routine maintenance during off peak times with advance notice and an automatic e-mail containing the new secret key will be sent to the TSP when this is done.

2.0 SERVICE SPECIFICATION

The service implements the following specifications:

1. Create a bill
2. Create multiple bills
3. Get bill information
4. Create a payment transaction
5. Register a tax payer
6. Get bill's invoice url
7. Search for a tax payer
8. Update payment redirect URL
9. Attach additional data to bill
10. Attach additional data to multiple bills

2.1 CREATE A BILL

2.1.1 API PATH

/api/ESBills/CreateESBill

2.1.2 REQUEST BODY

PROPERTY	DESCRIPTION	DATA TYPE	RE-QUIRED
engineCode	Engine code provided to the integrating party	String	Yes
identifier	Unique taxpayer's identifier used in existing system of integrating party	String	Yes
firstName	Names of the one paying the bill	String	Yes
middleName		String	Yes
lastName		String	Yes
telephone	Phone number of the one paying the bill	String	Yes
address	Address of the one paying the bill	String	Yes
esBillDetailsDto	Details of the bill being created	Array of ob-jects	Yes

For each object within the esBillDetailsDto array

PROPERTY	DESCRIPTION	DATA TYPE	RE-QUIRED
amount	Total amount for the bill detail	Decimal	Yes
mdasId	Id of the corresponding revenue head for the bill detail	Integer	Yes
recommend-edRate	(Optional)	Decimal	No
narration	Description of bill detail	String	Yes

2.1.3 RESPONSE BODY

PROPERTY	DESCRIPTION	DATA TYPE
bill	Information of created bill	Object
billItems	Bill details added to the bill	Array of objects
failedBillItems	Bill details that were NOT added to the bill	Array of objects

Within the bill object

PROPERTY	DESCRIPTION	DATA TYPE
billReference	Reference of created bill	String
payStatus	Bill's payment status. Will return 'Unpaid'	String
narration	Description of the bill	String
receipt	Receipt number. This will be blank as the bill is newly created	String

Within each object within the billItems array

PROPERTY	DESCRIPTION	DATA TYPE
billReference	Reference of bill the details is on	String
mdaId	Unique identifier of the revenue head	Integer
revenueCode	Unique code for the revenue head	String
revenueName	Name of revenue head	String
amount	Amount of the bill detail	Decimal
engineCode		String

Within each object within the failedBillItems array

PROPERTY	DESCRIPTION	DATA TYPE
billReference	Reference that bill detail failed to be added to	String
mdasId	The revenue head that is not attached to the integrating party's engine code	String
amount		Decimal

2.1.4 SAMPLE SCREENSHOTS

2.1.4.1 REQUEST

POST {{baseUrl}} /api/ESBills/CreateESBill

Params Authorization Headers (11) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary

```
1  {
2    "engineCode": "25477",
3    "identifier": "ABCD01234",
4    "firstName": "Yahya",
5    "middleName": "Umar",
6    "lastName": "Bindir",
7    "address": "PayKaduna HQ, Kaduna",
8    "telephone": "08010020300",
9    "esBillDetailsDto": [
10     {
11       "amount": 3400.50,
12       "mdasId": 138,
13       "narration": "Motor Vehicle Licenses"
14     },
15     {
16       "amount": 23750.99,
17       "mdasId": 125,
18       "narration": "PAYE Others"
19     }
20   ]
21 }
```

2.1.4.2 RESPONSE

POST {{baseUrl}}/api/ESBills/CreateESBill

Params Authorization Headers (12) Body Scripts Settings

Body Cookies Headers (5) Test Results

Pretty Raw Preview Visualize JSON


```
{
    "bill": {
        "billReference": "254770000215",
        "payStatus": "Unpaid",
        "narration": "FNKIR-24T-000015 MNKIR-24T-000015 LNKIR-24T-000015",
        "receipt": "-"
    },
    "billItems": [
        {
            "billReference": "254770000215",
            "mdaId": 138,
            "revenueCode": "12020630",
            "revenueHead": "Motor Vehicle Licences",
            "amount": 3400.50,
            "engineCode": null
        }
    ],
    "failedBillItems": [
        {
            "billReference": "254770000215",
            "mdasId": "PAYE Others",
            "amount": 2375.90
        }
    ]
}
```

2.2 CREATE MULTIPLE BILLS

2.2.1 API PATH

/api/ESBills/CreateBulkESBill

2.2.2 REQUEST BODY

PROPERTY	DESCRIPTION	DATA TYPE	REQUIRED
engineCode	Engine code provided to the integrating party	String	Yes
esBillDtos	All the bills being created	Array of objects	Yes

For each object within the esBillDtos array

PROPERTY	DESCRIPTION	DATA TYPE	REQUIRED
identifier	Unique taxpayer's identifier used in existing system of integrating party	String	Yes
firstName	Names of the one paying the bill	String	Yes
middleName		String	Yes
lastName		String	Yes
telephone	Phone number of the one paying the bill	String	Yes
address	Address of the one paying the bill	String	Yes
esBillDetailsDto	Details of the bill being created	Array of objects	Yes

For each object within the esBillDetailsDtos array

PROPERTY	DESCRIPTION	DATA TYPE	REQUIRED
amount	Total amount for the bill detail	Decimal	Yes
mdasId	Id of the corresponding revenue head for the bill detail	Integer	Yes
recommendedRate	(Optional)	Decimal	No
narration	Description of bill detail	String	Yes

2.2.3 RESPONSE BODY

PROPERTY	DESCRIPTION	DATA TYPE
	Array of created bills	Array of objects

For each object within the returned array

PROPERTY	DESCRIPTION	DATA TYPE
bill	Information of created bill	Object
billItems	Bill details added to the bill	Array of objects
failedBillItems	Bill details that were NOT added to the bill	Array of objects

Please see the response body of 2.1 CREATE A BILL to see what is contained within the billItems and failedBillItems properties.

2.2.4 SAMPLE SCREENSHOTS

2.2.4.1 REQUEST

POST {{baseUrl}} /api/ESBills/CreateBulkESBill

Params Authorization Headers (11) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary

```
{
  "engineCode": "25477",
  "esBillDtos": [
    {
      "identifier": "<string>",
      "firstName": "Yahya",
      "middleName": "Umar",
      "lastName": "Bindir",
      "address": "PayKaduna HQ, Kaduna",
      "telephone": "08010020300",
      "esBillDetailsDto": [
        {
        "amount": 3400.50,
        "mdasId": 138,
        "narration": "Motor Vehicle Licenses"
        }
      ]
    }
  ]
}
```

2.2.4.2 RESPONSE

POST {{baseUrl}}/api/ESBills/CreateBulkESBill

Params Authorization Headers (12) Body Scripts Settings

Body Cookies Headers (5) Test Results

Pretty Raw Preview Visualize JSON

```
[
    {
        "bill": {
            "billReference": "254770000216",
            "payStatus": "Unpaid",
            "narration": "FNKIR-24T-000015 MNKIR-24T-000015 LNKIR-24T-000015",
            "receipt": "-"
        },
        "billItems": [
            {
                "billReference": "254770000216",
                "mdaId": 138,
                "revenueCode": "12020630",
                "revenueHead": "Motor Vehicle Licences",
                "amount": 23456,
                "engineCode": null
            }
        ],
        "failedBillItems": []
    }
]
```

2.3 GET BILL INFORMATION

2.3.1 API PATH

/api/ESBills/GetBill?billreference=

2.3.2 REQUEST QUERY PARAMETERS

PARAMETER	DESCRIPTION	DATA TYPE	RE-QUIRED
billreference	The reference number generated when the bill was created	String	Yes

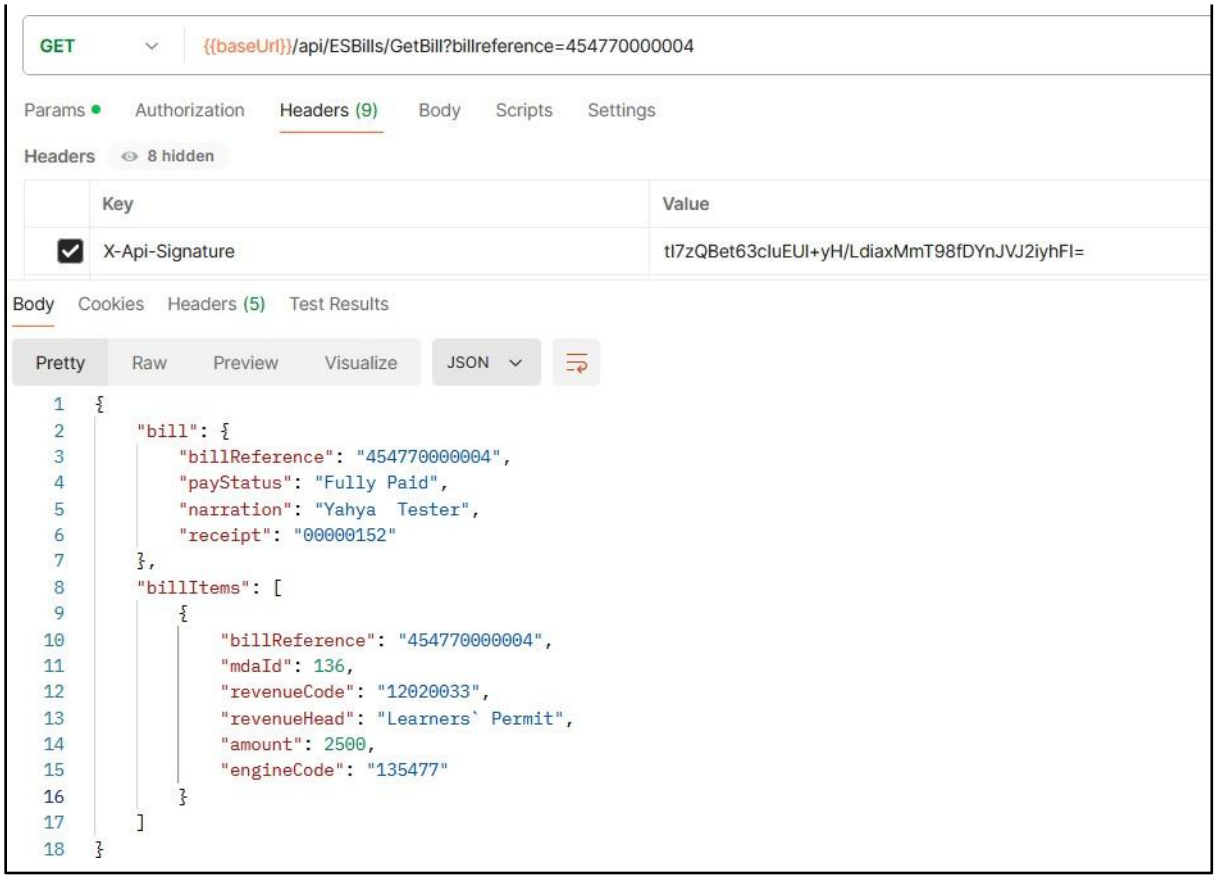
2.3.3 RESPONSE BODY

PROPERTY	DESCRIPTION	DATA TYPE
bill	Information of created bill	Object
billItems	Bill details added to the bill	Array of objects

Please see the response body of 2.1 CREATE A BILL to see what is contained within the bill and billItems properties

2.3.4 SAMPLE SCREENSHOTS

2.3.4.1 REQUEST & RESPONSE



2.4 CREATE A PAYMENT TRANSACTION

2.4.1 API PATH

/api/ESBills/CreateESTransaction

2.4.2 REQUEST BODY

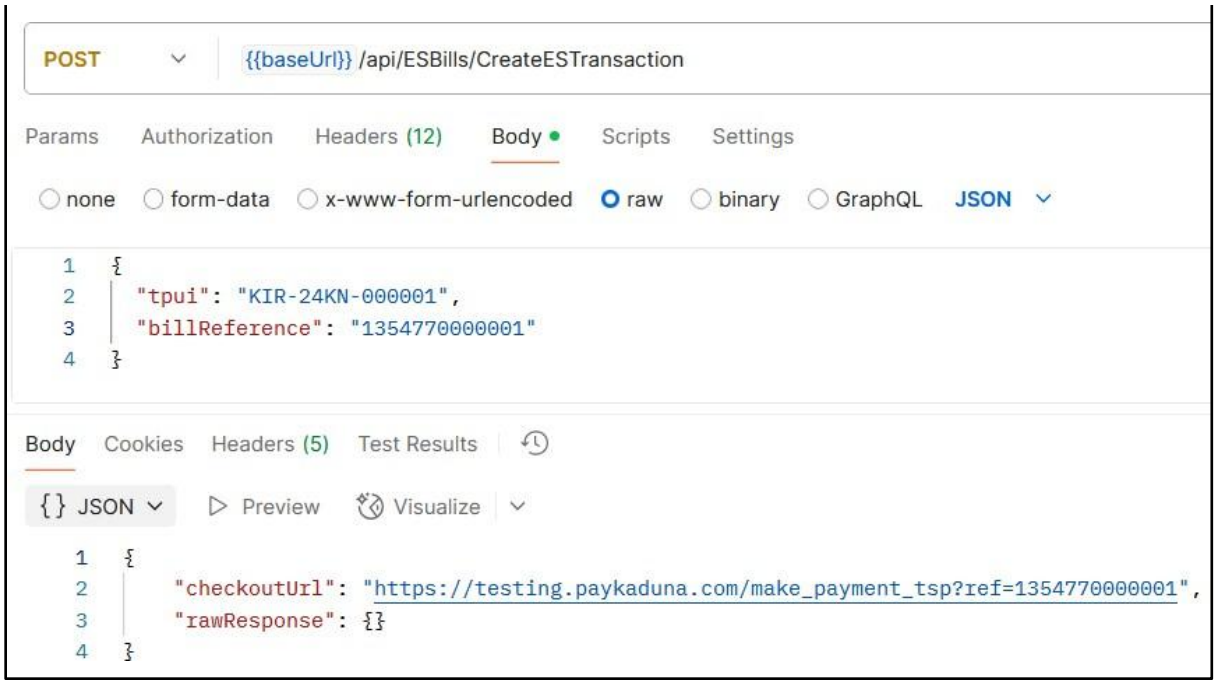
PROPERTY	DESCRIPTION	DATA TYPE	RE-QUIRED
tpui	A taxpayer's unique identifier. Used by PayKaduna	String	Yes
billReference	The reference number of the bill being paid	String	Yes

2.4.3 RESPONSE BODY

PROPERTY	DESCRIPTION	DATA TYPE
checkoutUrl	URL to the PayKaduna payment page	String
rawResponse	Any extra information required	Object

2.4.4 SAMPLE SCREENSHOTS

2.4.4.1 REQUEST & RESPONSE



2.5 REGISTER A TAX-PAYER

2.5.1 API PATH

/api/ESBills/RegisterTaxPayer

2.5.2 REQUEST BODY

PROPERTY	DESCRIPTION	DATA TYPE	REQUIRED
firstName		String	Yes
lastName		String	Yes
tin		String	Yes
email		String	Yes
phoneNumber		String	Yes
genderId		Integer	Yes
addressLine1	The address of the taxpayer	String	Yes
identifier	Unique taxpayer's identifier used in existing system of integrating party	String	Yes
userType		String	Yes
password		String	Yes
confirmPassword		String	Yes
middleName		String	No
rcNumber		String	No
industryId		Integer	No
numberOfStaff	Estimate of how many employees that have been employed	Integer	No
officeName		String	No
officeEmail		String	No
officePhoneNumber		String	No
officeAddressLine1		String	No
officeStateId	The state where the office's HQ stays	Integer	No
officeLgaId	The LGA where the office's HQ stays	Integer	No
taxStationId	The tax office for the registering tax payer	Integer	No

Supported values for the userType property are: "Individual", "Corporate"

N.B. If using the "Corporate" userType, please fill in the rcNumber, industryId, numberOfStaff, officeName, officeEmail, officePhoneNumber, officeAddressLine1, officeStateId, officeLgaId properties

2.5.3 RESPONSE BODY

PROPERTY	DESCRIPTION	DATA TYPE
userRegistrationId		String
userRegistration	Details of the taxpayer's registration on PayKaduna	Object
user	Basic info of taxpayer	Object
userType		String

The objects in the response are mostly what has been entered for the taxpayer's registration. However, the important new information can be found **within the userRegistration object**

PROPERTY	DESCRIPTION	DATA TYPE
tpui	The taxpayer's unique identifier on PayKaduna	String

2.5.4 SAMPLE SCREENSHOTS

2.5.4.1 REQUEST

POST {{baseUrl}} /api/ESBills/RegisterTaxPayer

Params Authorization Headers (12) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
{
  "firstName": "Yahya",
  "lastName": "Bindir",
  "middleName": "Testing",
  "identifier": "ABCDEF-10001",
  "tin": "0022345678",
  "taxStationId": 3,
  "genderId": 2,
  "addressLine1": "Plot 3, House 4",
  "email": "yahyatesting@email.com",
  "phoneNumber": "08100764536",
  "userType": "Individual",
  "password": "ApiTest123",
  "confirmPassword": "ApiTest123"
}
```

2.5.4.2 RESPONSE

POST {{baseUrl}}/api/ESBills/RegisterTaxPayer

Params Authorization Headers (12) Body Scripts Settings

Body Cookies Headers (6) Test Results

Pretty Raw Preview Visualize JSON


```
{
    "userRegistrationId": "11b0f11a-f39c-4b0f-9c1f-b404e6cd19a3",
    "userRegistration": {
        "id": "11b0f11a-f39c-4b0f-9c1f-b404e6cd19a3",
        "userId": "3c957caa-5b91-4464-a2a8-af39d1bc6ee4",
        "user": null,
        "fullName": "Yahya Testing Bindir",
        "tin": "0022345678",
        "tpui": "KIR-24BG-000006",
        "genderId": 2,
        "gender": null,
        "addressLine1": "Plot 3, House 4",
        "stateId": 19,
        "state": null,
        "lgaId": 1,
        "lga": null,
        "createdAt": "2024-09-20T22:06:40.806865Z",
        "modifiedAt": "2024-09-20T22:06:40.806865Z",
        "rcNumber": "-",
        "industryId": 1,
        "industry": null,
        "officeName": "-",
        "officeEmail": "-",
        "officePhoneNumber": "-",
        "officeAddressLine1": "-",
```


2.6 GET BILL'S INVOICE URL

2.6.1 API PATH

/api/ESBills/GetInvoiceUrl

2.6.2 REQUEST PARAMETERS

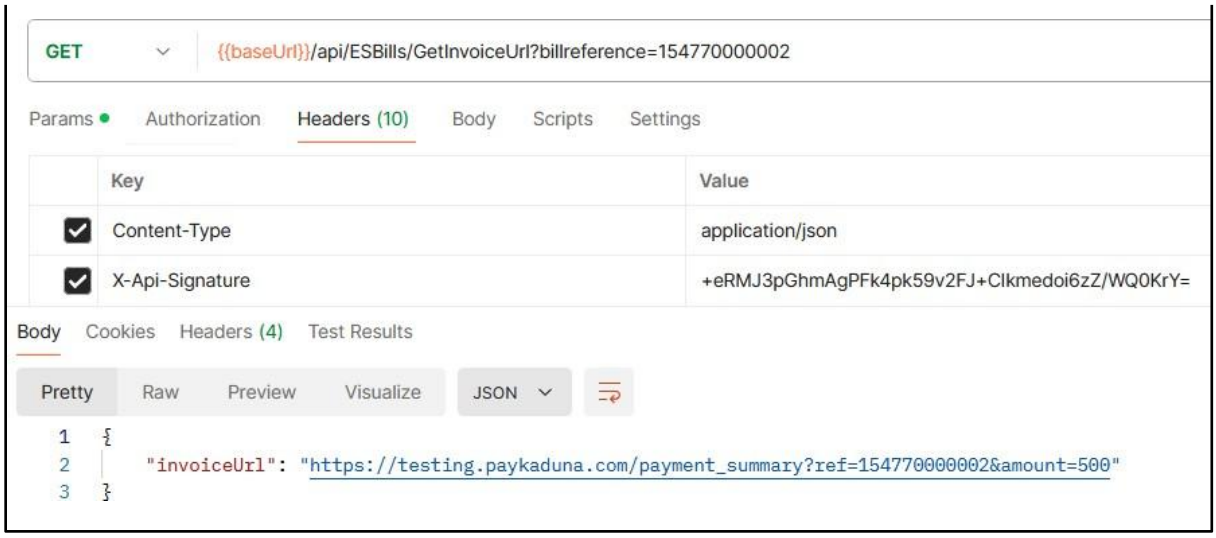
PARAMETER	DESCRIPTION	DATA TYPE	RE-QUIRED
billreference	The reference number generated when the bill was created	String	Yes

2.6.3 RESPONSE BODY

PARAMETER	DESCRIPTION	DATA TYPE
invoiceUrl	Link to the PayKaduna page where the taxpayer can view their invoice	String

2.6.4 SAMPLE SCREENSHOTS

2.6.4.1 REQUEST & RESPONSE



2.7 SEARCH TAX PAYER

2.7.1 API PATH

/api/ESBills/SearchTaxPayer?criteria=

2.7.2 REQUEST QUERY PARAMETERS

PARAMETER	DESCRIPTION	DATA TYPE	RE-QUIRED
criteria	The search term, minimum of 4 characters. This searches taxpayers by: i. Name ii. TPUI iii. TIN iv. NIN v. Phone number vi. Email	String	Yes

2.7.3 RESPONSE BODY

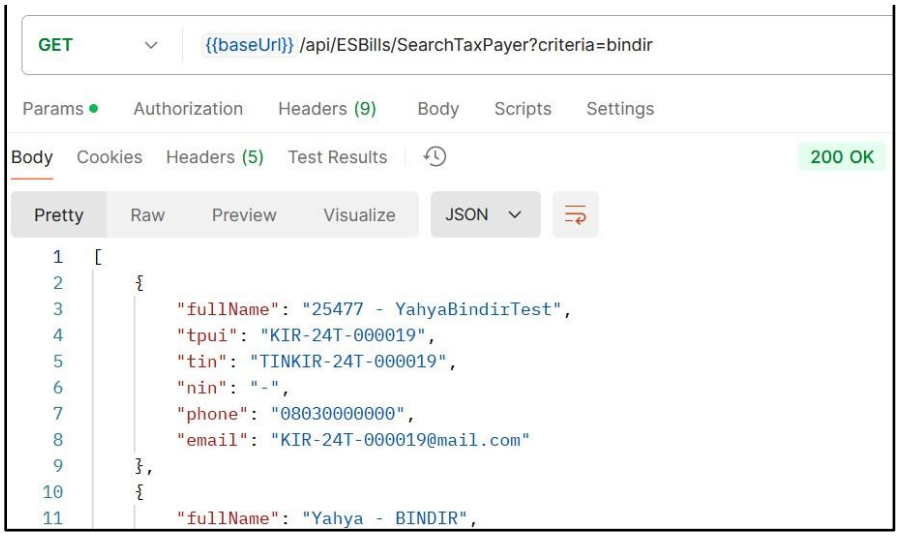
PROPERTY	DESCRIPTION	DATA TYPE
	An array of objects containing unpagi-nated search results that match the criteria provided	Array of objects
	Information of users	Object

Each object in the array contains the following information of the tax payer:

PROPERTY	DESCRIPTION	DATA TYPE
fullName		String
tpui		String
tin		String
nin		String
phone		String
email		String

2.7.4 SAMPLE SCREENSHOTS

2.7.4.1 REQUEST & RESPONSE



2.8 UPDATE PAYMENT REDIRECT URL

2.8.1 API PATH

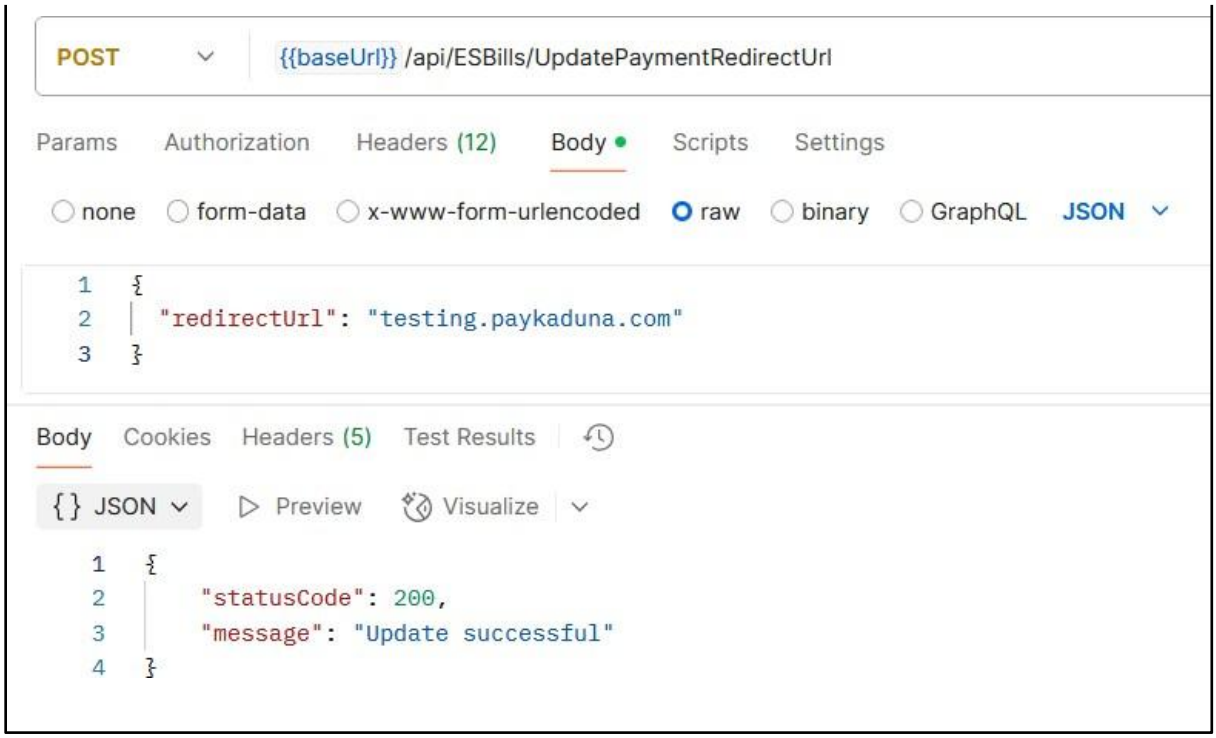
/api/ESBills/UpdatePaymentRedirectUrl

2.8.2 REQUEST QUERY PARAMETERS

PARAMETER	DESCRIPTION	DATA TYPE	RE-QUIRED
redirectUrl	This is the url taxpayers will be redirected to after payment has been made.	String	Yes

2.8.3 SAMPLE SCREENSHOTS

2.8.3.1 REQUEST & RESPONSE



2.9 ATTACH ADDITIONAL DATA TO BILL

2.9.1 API PATH

/api/ESBills/AttachAdditionalDataToBill

2.9.2 REQUEST BODY

PROPERTY	DESCRIPTION	DATA TYPE	REQUIRED
billReference	The PayKaduna bill reference	String	Yes
additionalData	Additional data to be attached to the given bill	Object	Yes

Note: The "additionalData" object has no set structure and is flexible for you to attach any custom attributes to the given bill.

2.9.3 RESPONSE BODY

PROPERTY	DESCRIPTION	DATA TYPE
billReference	The PayKaduna bill reference	String
tpui	The taxpayer's TPUI	String
payStatus	Payment status of bill	String
narration	Description of payment	String
receipt	Receipt number (if paid)	String
additionalData	Additional data that was attached	Object

2.9.4 SAMPLE SCREENSHOTS

2.9.4.1 REQUEST

POST {{baseUrl}} /api/ESBills/AttachAdditionalDataToBill

Params Authorization Headers (13) Body Scripts Set

none form-data x-www-form-urlencoded raw

```
{
  "billReference": "1354770000372",
  "additionalData": {
    "customAttribute": "attribute 1",
    "hiddenCost": 1000
  }
}
```

2.9.4.2 RESPONSE

POST {{baseUrl}} /api/ESBills/AttachAdditionalDataToBill

Params Authorization Headers (13) Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Body Cookies (1) Headers (5) Test Results 200 OK

{} JSON Preview Visualize

```
{
    "billReference": "1354770000372",
    "tpui": "KIR-25T-000114",
    "payStatus": "Unpaid",
    "narration": "Proxy Bill MDA User",
    "receipt": "-",
    "additionalData": {
        "hiddenCost": 1000,
        "customAttribute": "attribute 1"
    }
}
```

2.10 ATTACH ADDITIONAL DATA TO MULTIPLE BILLS

2.10.1 API PATH

/api/ESBills/BulkAttachAdditionalDataToBill

2.10.2 REQUEST BODY

PROPERTY	DESCRIPTION	DATA TYPE	REQUIRED
bills	The bills that will have additional data	Array of Objects	Yes

Note: Each object in the bills array will have the same structure as described in 2.9.2

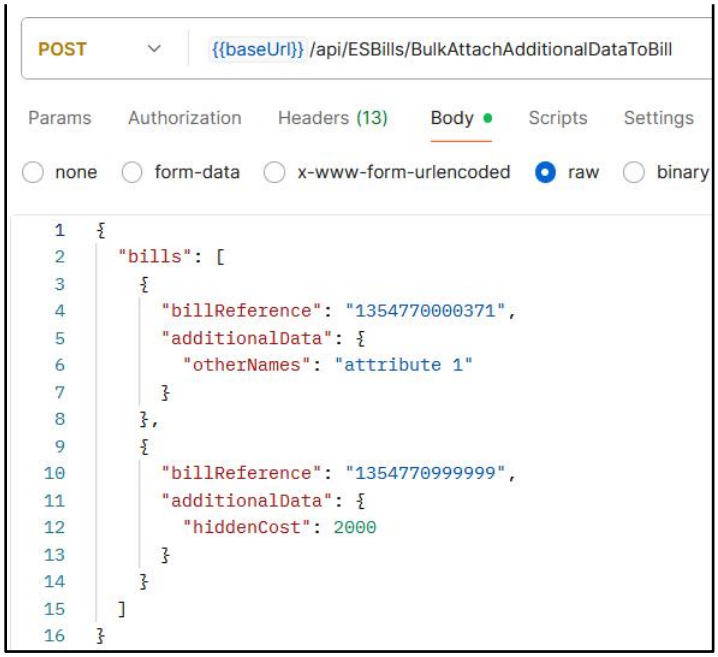
2.10.3 RESPONSE BODY

PROPERTY	DESCRIPTION	DATA TYPE
success		Bool
errors	List of error messages	Array of Strings
bills	The bills that had attached	Array of Objects

Note: Each object in the bills array will have the same structure as described in 2.9.3

2.10.4 SAMPLE SCREENSHOTS

2.10.4.1 REQUEST



2.10.4.2 RESPONSE

POST {{baseUrl}} /api/ESBills/BulkAttachAdditionalDataToBill

Params Authorization Headers (13) Body Scripts Settings

Body Cookies (1) Headers (5) Test Results

{} JSON Preview Visualize


```
{
    "success": false,
    "errors": [
        "Bill not found using 1354770999999"
    ],
    "bills": [
        {
            "billReference": "1354770000371",
            "tpui": "KIR-24CK-000003",
            "payStatus": "Unpaid",
            "narration": "Proxy Bill Tax Office User",
            "receipt": "-",
            "additionalData": {
                "otherNames": "attribute 1"
            }
        }
    ]
}
```


3.0 LOOKUP TABLES

3.1 STATES

ID	NAME
19	Kaduna

3.2 LGAS

ID	NAME
1	Birnin Gwari
2	Chikun
3	Giwa
4	Igabi
5	Ikara
6	Jaba
7	Jema'a
8	Kachia
9	Kaduna North
10	Kaduna South
11	Kagarko
12	Kajuru
13	Kaura
14	Kauru
15	Kubau
16	Kudan

17	Lere
18	Makarfi
19	Sabon Gari
20	Sanga
21	Soba
22	Zangon Kataf
23	Zaria

3.3 GENDERS

ID	NAME
1	Female
2	Male
3	Not Specified

3.4 TAX STATIONS

ID	NAME
2	Kakuri East
3	Tudun Wada
4	Kakuri West
5	Kafanchan
6	Zaria
7	Kawo
8	Doka East
9	Doka West
10	Samaru
11	Zonkwa
12	Kachia
13	Jere
14	Birnin Gwari
15	Turunku
16	Giwa
17	Makarfi
18	Ikara
19	Soba
20	Saminaka
21	Kaura
22	Kauru
23	Kudan
24	Kajuru
25	Kubau
26	Gwantu
27	Rigasa Mini
28	Sabo Mini
29	Oriakpata Mini
30	Sheikh Gumi Market
31	Kwoi
32	Sabon Gari
33	Abuja
34	CMR MLA
35	Obasanjo House MLA
36	FRSC MLA

3.5 INDUSTRIES

ID	NAME
1	Agriculture
2	Oil and Gas
3	Banking and Finance
4	Telecommunications
5	Manufacturing
6	Real Estate
7	Construction
8	Healthcare
9	Information Technology (IT)
10	Education
11	Transportation and Logistics
12	Hospitality and Tourism
13	Retail and Wholesale
14	Renewable Energy
15	Mining and Quarrying
16	Media and Entertainment
17	Insurance
18	Textile and Garment Industry
19	Automobile Industry